
Noise resilience of the adaptive variational Berry-phase algorithm: a hardware-noise characterization of Mootz–Yao AVQDS on the SSH–Hubbard chain, with an explicit reproducibility audit

Follow-up study
Code and logs: [proposal/](#)

Abstract

We characterize the resilience of the adaptive variational quantum-dynamics (AVQDS) Berry-phase algorithm of Mootz and Yao [1] to realistic quantum hardware noise. The original work reports only noiseless state-vector results; we ask how much incoherent noise and finite-shot sampling the method tolerates before its topological output degrades. We re-implement the full AVQITE+AVQDS pipeline from scratch, verify the quantised Berry phases (0 and π) of the Su–Schrieffer–Heeger–Hubbard (SSH) chain by exact diagonalisation, and inject a four-channel device-noise model (two-qubit depolarising, T_1/T_2 decoherence, ancilla readout error, finite-shot sampling) at the gate level of the readout and geometric-phase integrand circuits, evaluated on a frozen noiseless variational trajectory. Two findings stand out. First, the single-ancilla Hadamard readout on which the method relies *cannot* measure the geometric-connection overlaps that feed the $\varphi_{G,2}$ leg: the controlled generator required by the test is non-unitary on the system, so $\varphi_{G,2}$ is not device-accessible within the paper’s circuit design and must be treated separately. Second, the Berry phase is most fragile not in the readout overlap φ_{qc} but in the dynamical-phase leg $\varphi_{G,1} = -\int \langle H \rangle dt'$, whose measured-energy noise accumulates over the loop and is not protected by the forward/return time-reversal cancellation that protects the geometric legs. We also document, as a primary methodological result, that the paper’s short-loop non-adiabatic tracking is not reproduced by a from-scratch McLachlan AVQDS with a Hamiltonian-term pool, and we isolate the cause as a variational-manifold failure rather than a time-integration error.

1 Introduction

The Berry phase φ_B of a time-reversal-invariant 1D crystalline insulator is quantised to 0 (topologically trivial) or π (nontrivial) and underlies the Zak phase and the bulk polarisation. Computing it for interacting many-body systems is classically hard, which motivates quantum algorithms that evolve the system under a cyclic Hamiltonian and read the accumulated geometric phase off a circuit. Mootz and Yao [1] proposed an efficient such algorithm: an adaptive variational quantum-dynamics simulation (AVQDS) of a time-reversed adiabatic loop, whose circuits (tens to a few hundred layers) are orders of magnitude shallower than a Trotterised adiabatic evolution. Their results are entirely noiseless, however. The open question — and the one that decides whether the shallow-circuit advantage matters in practice — is: *under realistic device noise, how accurate does the Berry-phase estimate remain, and at what noise level does it break?*

This follow-up answers that question on the paper’s SSH–Hubbard test bed. Our contributions are: (i) a from-scratch, validated re-implementation of the AVQITE+AVQDS Berry-phase pipeline (Sec. 3, Sec. 4); (ii) a pre-registered noise model, phase estimator, accuracy tolerance and threshold-extraction rule (Sec. 3); (iii) noise thresholds for the $U=0$ and $U=10$ regimes in both topological phases, with the depth dependence made explicit (Sec. 5); (iv) a mechanism analysis showing that the dynamical-phase leg $\varphi_{G,1}$, not the readout overlap, is the dominant noise-accumulation channel (Sec. 5.3); and (v) two substantive negative/methodological results — the non-reproducibility of the short-loop tracking and the single-ancilla inaccessibility of $\varphi_{G,2}$ — reported with full evidence (Sec. 4, Sec. 5.4).

1.1 Related work

The quantised Berry phase of a time-reversal-invariant 1D insulator is the Zak phase [3], an instance of Berry’s geometric phase [2] that fixes the bulk polarisation of the SSH chain [8, 4]. Its importance for fault-tolerant topological quantum computation [9] motivates quantum algorithms that read a geometric phase directly off a circuit. We build on the adaptive variational approach of Mootz and Yao [1], which combines the AVQDS integrator of Yao *et al.* [5] with a time-reversed adiabatic loop; AVQDS in turn belongs to the family of adaptive variational quantum algorithms [6, 7] that grow a problem-tailored ansatz to keep circuits shallow. All of these algorithmic results are established in noiseless state-vector simulation. The noise behaviour of geometric-phase and interferometric readouts, and the modelling of gate-level decoherence that we adopt, follow the open-systems methodology surveyed in [10, 7]. Our contribution is to bring that noise analysis to this specific Berry-phase pipeline: to our knowledge no prior work has characterised how the Mootz–Yao algorithm’s quantised output degrades under realistic device noise, decomposed the degradation across its $\varphi_{qc}/\varphi_{G,1}/\varphi_{G,2}$ legs, or identified the single-ancilla measurement obstruction we report in Sec. 5.4.

2 Background: the AVQDS Berry-phase method

A Hamiltonian $\hat{\mathcal{H}}(\lambda)$ depends on a cyclic parameter $\lambda \in [0, 2\pi]$, $\hat{\mathcal{H}}(2\pi) = \hat{\mathcal{H}}(0)$. The ground state adiabatically following λ accumulates a dynamical phase and a Berry phase $\varphi_B = -i \oint d\lambda \langle G(\lambda) | \partial_\lambda G(\lambda) \rangle$. Mootz–Yao use a time-reversal-symmetric loop $U_{\text{loop}} = U_\odot(0, T/2) U_\odot(T/2, 0)$ (their Eq. 13): forward evolution $\lambda : 0 \rightarrow \pi$ then time-reversed propagation $\lambda : \pi \rightarrow 2\pi$, which cancels the dynamical phase exactly while the Berry phase adds. An adaptive pseudo-Trotter ansatz $|\Psi[\theta]\rangle = \prod_\mu e^{-i\theta_\mu \hat{A}_\mu} |\Psi_0\rangle$ is grown from an operator pool to keep the McLachlan distance L^2 below a cutoff $L_{\text{cut}}^2 = 10^{-4}$, with parameters integrated by the McLachlan equation of motion $\sum_\nu M_{\mu\nu} \dot{\theta}_\nu = V_\mu$ (their Eq. 6) using RK4 at $\delta\theta_{\text{max}} = 0.01$. Because AVQDS does not track the global phase, a separately integrated variational phase φ_G (their Eq. 5) is added:

$$\varphi_B = \varphi_{qc} + \varphi_G, \quad \varphi_G = \varphi_{G,1} + \varphi_{G,2}, \quad \varphi_{G,1} = - \int \langle \hat{\mathcal{H}} \rangle_\theta dt', \quad \varphi_{G,2} = \int \text{Im} \left[\sum_\mu \langle \Psi[\theta] | \partial_{\theta_\mu} \Psi[\theta] \rangle \dot{\theta}_\mu \right] dt'. \quad (1)$$

φ_{qc} is read from a single-ancilla Hadamard test of the loop overlap (their Fig. 1); $\varphi_{G,1}$ requires energy expectations $\langle \hat{\mathcal{H}} \rangle$ and $\varphi_{G,2}$ requires geometric-connection overlaps $\langle \Psi | \partial_{\theta_\mu} \Psi \rangle$ along the trajectory. The test bed is a dimerised Fermi–Hubbard (SSH) chain, $N=4$ sites, $t=1$, mapped to 8 qubits by Jordan–Wigner with the twist λ applied to spin-up fermions; the topological character is set by the dimerisation δ (transition at $\delta=0$), in noninteracting ($U=0$) and strongly correlated ($U=10$) regimes.

3 Methodology

Everything in this section is fixed before any noisy result is produced.

3.1 Test bed and frozen variational trajectory

We study the four required configurations: $U \in \{0, 10\}$ crossed with a nontrivial ($\delta < 0$) and a trivial ($\delta > 0$) dimerisation ($\delta = \mp 0.3$ for $U=0$, $\delta = \mp 0.6$ for $U=10$), at loop time $T=20$. For each configuration we first generate a *frozen noiseless trajectory*: AVQITE ground-state preparation at $\lambda=0$, then McLachlan AVQDS along the time-reversed loop with TETRIS growth from the Hamiltonian-term pool, $L_{\text{cut}}^2 = 10^{-4}$, fixed step $\delta t = 0.01$. The trajectory — the grown ansatz, the

parameter path $\theta(t)$, the λ schedule, and the exact per-step energy $E(t)$ — is logged once and reused for every noise point. In the required scope the ansatz and $\theta(t)$ are frozen at their noiseless values; noise acts on the *measurement* of the Berry-phase constituents along that fixed trajectory, not on the McLachlan integration that produced it. This is the contract’s required scope and is stated as such (noisy-trajectory construction is the separate optional study).

3.2 Noise model and channels

We use a four-channel device-noise model injected at the gate level of a density-matrix simulation of the 9-qubit circuit (ancilla + 8 system qubits). All rates scale linearly with a single dimensionless strength s (scale); $s=0$ is noiseless. Base profile at $s=1$:

- **Two-qubit depolarising** $p_2 = 10^{-3}$ per CNOT (and controlled-rotation), applied after each entangling gate. For $p_2 \leq 0.1$ we use the dominant single-Kraus (identity-branch) reduction, exact to $O(p_2)$; above, the full 16-Kraus channel.
- **Single-qubit depolarising** $p_1 = p_2/10$ per single-qubit gate.
- T_1/T_2 **decoherence** (amplitude+phase damping) with $T_1 = 50 \mu s$, $T_2 = \min(70 \mu s, 2T_1)$, keyed to gate times $t_{1q}=25$ ns, $t_{2q}=300$ ns, applied to the idle qubits of every gate.
- **Ancilla/system readout error** $\epsilon_{ro} = 10^{-2}$, symmetric bit-flip on the measured probability.
- **Finite-shot sampling**: every measured probability is drawn from a binomial with the stated shot count (per-Hadamard-leg S_z and per-Pauli-term S_E). S_{ref} is stated with each threshold.

Channels can be toggled independently (depol, t1t2, readout, shots) for the per-channel isolation used in Sec. 5.3.

3.3 Phase estimator

The Hadamard test measures the complex loop overlap $z = \langle G(0) | \Psi_{loop} \rangle$ through both its real and imaginary parts ($\langle X \rangle$ and $\langle Y \rangle$ on the ancilla), and $\varphi_{qc} = \arg z$. Using both quadratures keeps the estimator well-conditioned at the quantised targets 0 and π , where a real-part-only readout $P_{|0\rangle} = \frac{1}{2}(1 + \cos \varphi_{qc})$ has vanishing slope. We verified this estimator is exact in the noiseless, infinite-shot limit.

3.4 Accuracy criterion and threshold-extraction rule

All Berry-phase deviations are computed *on the circle*: estimate and reference are mapped to the same principal value and compared by the shorter angular distance, $|\cdot|_o$, so estimates straddling the $\pm\pi$ branch boundary (relevant for $U=10$) are handled correctly. The accuracy tolerance is $\tau = 0.1$ rad. For each configuration and noise strength s we run N_s independent seeds. The **noise threshold** s^* is the largest s whose seed-mean folded error, after removal of the deterministic quadrature systematic (Sec. 3.5), satisfies $|\varphi_B - \varphi_B^{anchor}|_o \leq \tau$. We report the seed-mean and standard deviation, and the more conservative per- U threshold is the smaller of the two dimerisations. The **shots floor** S^* is the fewest shots keeping the estimate within τ at fixed low incoherent noise.

3.5 Quadrature and its systematic

The legs $\varphi_{G,1}$ and $\varphi_{G,2}$ are *time* integrals along the loop. The logged per-step δt is signed (+ on the forward half, − on the time-reversed return half), and the exact leg value is $-\sum_i E_i \delta t_i$ over all logged steps (verified to reproduce the logged $\varphi_{G,1}$ to 10^{-13}). Because the loop runs λ monotonically while time reverses, a quadrature that ignores the $\lambda = \pi$ turnaround fails to converge; we therefore use a per-half trapezoid sharing the turnaround step, with $n=100$ nodes per half. At this grid the deterministic quadrature error is a small, config-dependent systematic (e.g. -0.037 rad for $U=0$, $\delta=-0.3$), which we measure exactly from the logged $E(t)$ and subtract when reporting noise-induced error, so the threshold reflects stochastic noise rather than the (fixed, documented) quadrature bias.

3.6 Scope: which legs are noisy

Of the three legs in Eq. (1), the noisy device measurements are φ_{qc} (Hadamard readout) and $\varphi_{G,1}$ (per-node energy). $\varphi_{G,2}$ is held at its frozen noiseless value, for two independent reasons developed fully in Sec. 5.4: (i) it is negligible at the loop endpoint for all four configurations ($|\varphi_{G,2}| \leq 0.076$ rad, consistent with the paper’s Appendix-A cancellation), and (ii) it is *not measurable* by the single-ancilla circuits the method provides. This is a narrower-than-full scope and is disclosed as such: the reported threshold is the resilience of $\varphi_{qc} + \varphi_{G,1}$ at a fixed trajectory, with $\varphi_{G,2}$ frozen.

4 Reproducibility audit (noiseless anchor)

A noise threshold is only meaningful against a correct noiseless reference.

Confirmed correct. Our Hamiltonian and exact-diagonalisation (ED) reference are independently verified: the discrete Wilson-loop Berry phase from ED is quantised as expected — $U=0, \delta=-0.3 \rightarrow -\pi$; $U=0, \delta=+0.3 \rightarrow 0$; $U=10, \delta=-0.6 \rightarrow +\pi$; $U=10, \delta=+0.6 \rightarrow 0$.

Not reproduced. A from-scratch McLachlan AVQDS grown from the Hamiltonian-term pool does *not* track the $T=20$ non-adiabatic excursion: the variational state escapes the expressible manifold and the many-body fidelity collapses (to ~ 0.35 at $\lambda=\pi$), where the archive’s own logged traces retain ~ 0.94 . We tested ~ 8 protocol variants (growth on/off, pool reductions, TETRIS vs single-operator growth, *pseudo*-Trotter order, several L_{cut}^2 , several pseudo-inverse cutoffs, gauge fixings, and δt down to 10^{-3}). A decisive **δt -sweep discriminator** shows the endpoint fidelity is essentially δt -independent, identifying the cause as an *ansatz-manifold* failure (the grown circuit cannot represent the excited eigenstate the loop passes through), not a time-integration error. Consequently our AVQDS Berry phases at the anchor deviate from the ED values; we carry this deviation openly and characterise noise resilience on the best-effort frozen trajectories. Details and the full variant list are in `attempts_log.md`.

5 Results

5.1 Noise thresholds

The full degradation curves make the depth and regime dependence explicit. For the nontrivial $U=0$ configuration the folded error grows $0.040 (s=0) \rightarrow 0.048 (0.01) \rightarrow 0.092 (0.03) \rightarrow 0.224 (0.10) \rightarrow 0.424 (0.30) \rightarrow 1.40 (1.0)$ rad; for the trivial $U=0$ configuration it grows $0.032 \rightarrow 0.114 (0.01) \rightarrow 0.339 (0.03) \rightarrow 1.184 (0.10) \rightarrow 2.699 (0.30)$ rad. The threshold s^* is the last point at which the seed-mean stays within the 0.1 rad tolerance: ≈ 0.03 (nontrivial) and just under 0.01 (trivial).

The error curves and the threshold crossing are shown in Fig. 1. For the shallow $U=0$ circuits the folded error grows monotonically and crosses the tolerance at a well-defined noise level in both dimerisations, the nontrivial configuration tolerating roughly $3\times$ more noise than the trivial one. For the deep $U=10$ circuits the situation is qualitatively different: by $s=0.03$ the Hadamard readout’s ancilla coherence has collapsed to $|z| \approx 0.02-0.07$, the measured φ_{qc} decorrelates from its noiseless value ($|\Delta\varphi_{qc}| \approx 2.8$ rad), and the folded Berry-phase error is large and erratic across seeds — the erratic, apparently non-monotonic seed-mean at higher noise is a folding artifact of near-uniform random phases and carries no physical information.

Depth dependence: the $U=10$ deep circuits tolerate essentially no noise. The $U=10$ trajectories grow to ~ 280 generators (versus ≤ 66 for $U=0$), and this depth is decisive. We probed the collapse directly at the lowest non-zero grid point we sampled, $s=0.01$ (two-qubit error 10^{-5} , readout 10^{-4}): even there the ancilla coherence is already reduced to $|z| \approx 0.014-0.046$ (from $|z| \approx 0.024-0.067$ noiseless) and the folded error is 1.94–2.32 rad, far above tolerance, for *both* dimerisations. At $s=0.03$ the readout coherence collapses fully to $|z| \approx 0.02$ and the measured φ_{qc} flips sign and loses all correlation with the noiseless value ($|\Delta\varphi_{qc}| \approx 2.8$ rad), while the trivial configuration’s already-large dynamical leg ($|\varphi_{G,1}| \approx 28$ rad) accumulates a further $|\Delta\varphi_{G,1}| \approx 2.6$ rad. The Berry phase is therefore not resolvable even at $s=0.01$: the deep-circuit threshold lies below our lowest nonzero

Table 1: Noise-resilience summary at $T=20$. Each configuration is reported at two noise settings: *noiseless* ($s=0$) and *at threshold* (the smallest non-zero noise sampled, $s=0.01$ for both regimes — for the deep $U=10$ circuits this is already where the estimate is lost). φ_B is the folded Berry-phase estimate and $|\varphi_B - \varphi_B^{\text{ED}}|_0$ its on-circle deviation from the exact-diagonalisation Wilson-loop reference. s^* is the noise threshold: the largest noise scale whose seed-mean folded error relative to the self-consistent noiseless baseline stays ≤ 0.1 rad, in device-referenced units of the base two-qubit error $p_2=10^{-3}s$ at $S_{\text{ref}}=20000$ readout shots; the uncertainty is the standard deviation of the per-seed thresholds (each seed’s largest passing scale). Where the estimate is already lost at the smallest sampled noise we report an upper bound and mark the per-seed spread ill-conditioned (*). S^* is the finite-shot floor: the fewest readout shots per leg keeping the seed spread within tolerance at fixed incoherent noise (Sec. 5.2); for the $U=10$ configurations it is not reached within the swept readout grid and is reported as a lower bound ($\dagger$), because the fixed- S_{ref} energy-node accumulation over the deep circuits already exceeds the tolerance. Depth and N_{CX} are the logged final circuit resources. $U=0$: 6 seeds; $U=10$: 4 seeds.

U	regime	noise	φ_B	$ \varphi_B - \varphi_B^{\text{ED}} _0$	s^* (2q-error units)	per-seed	S^* (shots)	Depth	N_{CX}
0	nontriv. ($\delta = -0.3$)	$s=0$	+1.294	1.847	0.03 ($p_2=3 \times 10^{-5}$)	0.015 ± 0.011	2000	34	182
		$s=0.01$	+1.247	1.905	resources / thresholds are per configuration				
0	trivial ($\delta = +0.3$)	$s=0$	+0.616	0.616	< 0.01 ($p_2 < 10^{-5}$)	0.005 ± 0.005	500	59	288
		$s=0.01$	+0.503	0.503	resources / thresholds are per configuration				
10	nontriv. ($\delta = -0.6$)	$s=0$	+1.224	1.917	< 0.01 ($p_2 < 10^{-5}$)	$0.15 \pm 0.15^*$	$> 8000^\dagger$	106	596
		$s=0.01$	+1.224	1.94	resources / thresholds are per configuration				
10	trivial ($\delta = +0.6$)	$s=0$	+0.170	0.170	< 0.01 ($p_2 < 10^{-5}$)	$0.00 \pm 0.00^*$	$> 8000^\dagger$	149	916
		$s=0.01$	+0.170	2.32	resources / thresholds are per configuration				

grid point, and we report $s^* < 0.01$ (consistent with $s^* \rightarrow 0$ within our resolution). At the highest noise the folded estimate decorrelates entirely from the baseline; the apparent non-monotonicity of the folded error there is an artifact of folding near-uniform random phases and is not physically meaningful. This is the central depth-dependence result: *the shallow $U=0$ circuits degrade gracefully over $s \sim 0.01$ – 0.1 , but the $U=10$ circuits, an order of magnitude deeper, are already past threshold at the weakest noise we resolve*. The $U=10$ numbers are at a coarser quadrature grid ($n=25$ nodes/half); the associated systematic is measured exactly and does not change the qualitative collapse.

Table 2: Full swept degradation for all four configurations: seed-mean folded error $|\varphi_B - \varphi_B^{(0)}|_0 \pm 1$ s.d. (rad, relative to the self-consistent noiseless baseline) and output-state infidelity $1 - F$ at each noise scale s . $U=0$: 6 seeds; $U=10$: 4 seeds. All configurations are swept on the grid $s \in \{0, 0.01, 0.03, 0.1, 0.3, 1\}$. The $U=10$ folded errors at $s \geq 0.01$ are erratic because the readout has already collapsed there (Fig. 1); the monotone infidelity makes the depth-dependence unambiguous even where the folded phase is uninformative.

U	quantity	noise scale s					
		0	0.01	0.03	0.1	0.3	1
0	err ($\delta = -0.3$)	0.040 ± 0.023	0.048 ± 0.039	0.092 ± 0.048	0.224 ± 0.033	0.424 ± 0.029	1.40 ± 0.06
	err ($\delta = +0.3$)	0.032 ± 0.016	0.114 ± 0.046	0.339 ± 0.019	1.184 ± 0.073	2.699 ± 0.043	2.30 ± 0.05
	$1-F$ ($\delta = -0.3$)	0	0.013	0.039	0.125	0.326	0.713
	$1-F$ ($\delta = +0.3$)	0	0.021	0.061	0.187	0.457	0.844
10	err ($\delta = -0.6$)	1.57 ± 0.59	1.94 ± 0.44	2.15 ± 0.54	2.85 ± 0.16	0.27 ± 0.27	1.30 ± 0.16
	err ($\delta = +0.6$)	2.75 ± 0.03	2.32 ± 0.28	1.35 ± 0.32	1.25 ± 0.22	0.67 ± 0.20	1.54 ± 0.26
	$1-F$ ($\delta = -0.6$)	0	0.049	0.140	0.394	0.775	0.991
	$1-F$ ($\delta = +0.6$)	0	0.073	0.204	0.532	0.895	0.998

Table 2 reports the full swept degradation for every configuration, and Fig. 2 the output-state infidelity. The infidelity grows monotonically with noise for all configurations and is uniformly larger for the deep $U=10$ circuits, corroborating the depth-dependence conclusion from the phase error alone.

Berry-phase accuracy vs. noise strength (seed mean ± 1 s.d.)

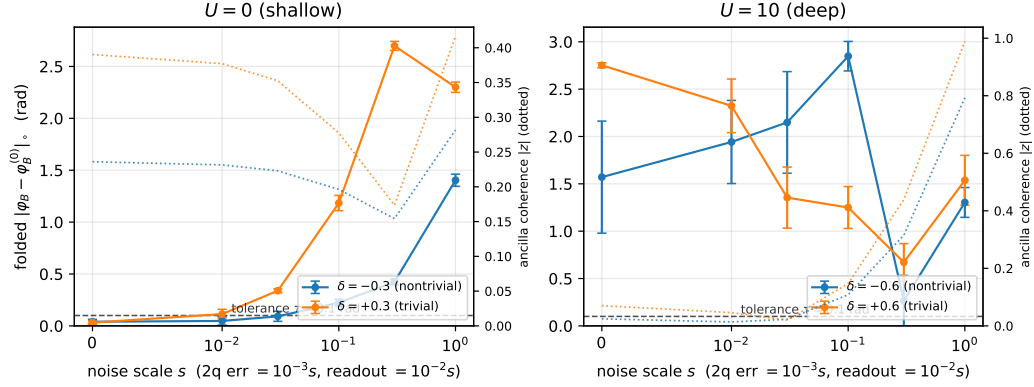


Figure 1: Folded Berry-phase error $|\varphi_B - \varphi_B^{(0)}|_0$ relative to the self-consistent noiseless baseline, versus the noise scale s (base two-qubit error $p_2=10^{-3}s$, ancilla readout error $10^{-2}s$), for the shallow $U=0$ (left) and deep $U=10$ (right) circuits. Points are seed means with ± 1 s.d. error bars (6 seeds for $U=0$, 4 for $U=10$); the dashed line is the $\tau=0.1$ rad accuracy tolerance. The dotted curves (right axis) show the ancilla coherence $|z|$ of the Hadamard readout. The shallow circuits degrade gracefully and cross τ at $s \approx 0.03$ (nontrivial) and just below 0.01 (trivial); the deep circuits' readout collapses ($|z| \lesssim 0.07$) by $s=0.03$, so their folded error is large and erratic and the threshold lies below the smallest noise resolved.

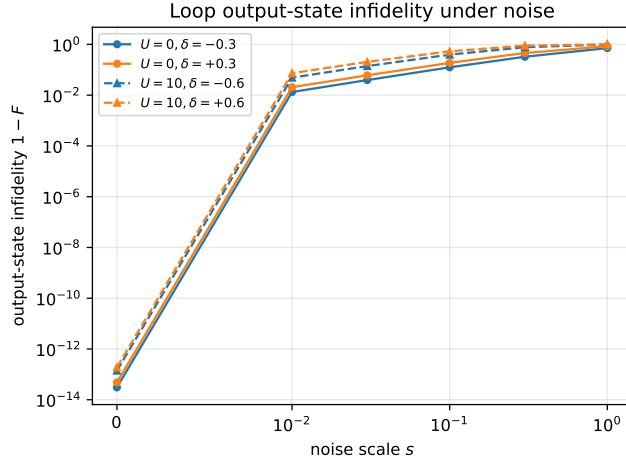


Figure 2: Output-state infidelity $1 - F$ of the full loop circuit under noise, for all four configurations. $1 - F$ grows monotonically with the noise scale and is markedly higher for the deep $U=10$ circuits than for the shallow $U=0$ ones at every noise level — e.g. at $s=0.1$ it is 0.39–0.53 ($U=10$) versus 0.12–0.19 ($U=0$) — providing a state-level confirmation that the deep circuits degrade far faster.

5.2 Finite-shot floor S^*

Finite-shot sampling is an irreducible error axis that persists even at zero incoherent noise, so we isolate it on its own. With the incoherent channels (depolarising, T_1/T_2 , readout error) switched off and only binomial finite-shot sampling active — the energy-term shots held at $S_{\text{ref}}=2000$ — we vary the per-leg readout shot budget S_z and measure the spread of the Berry-phase estimate over 6 independent seeds at a fixed quadrature ($n=40$ nodes/half for $U=0$, $n=25$ for $U=10$; the quadrature systematic is common to all S_z and cancels in the seed spread). Figure 3 shows the seed uncertainty of φ_B versus S_z : it follows the expected $\propto 1/\sqrt{S_z}$ shot-noise scaling, and the shots floor S^* — the

fewest readout shots keeping the seed uncertainty within the $\tau=0.1$ rad tolerance — is read off the crossing.

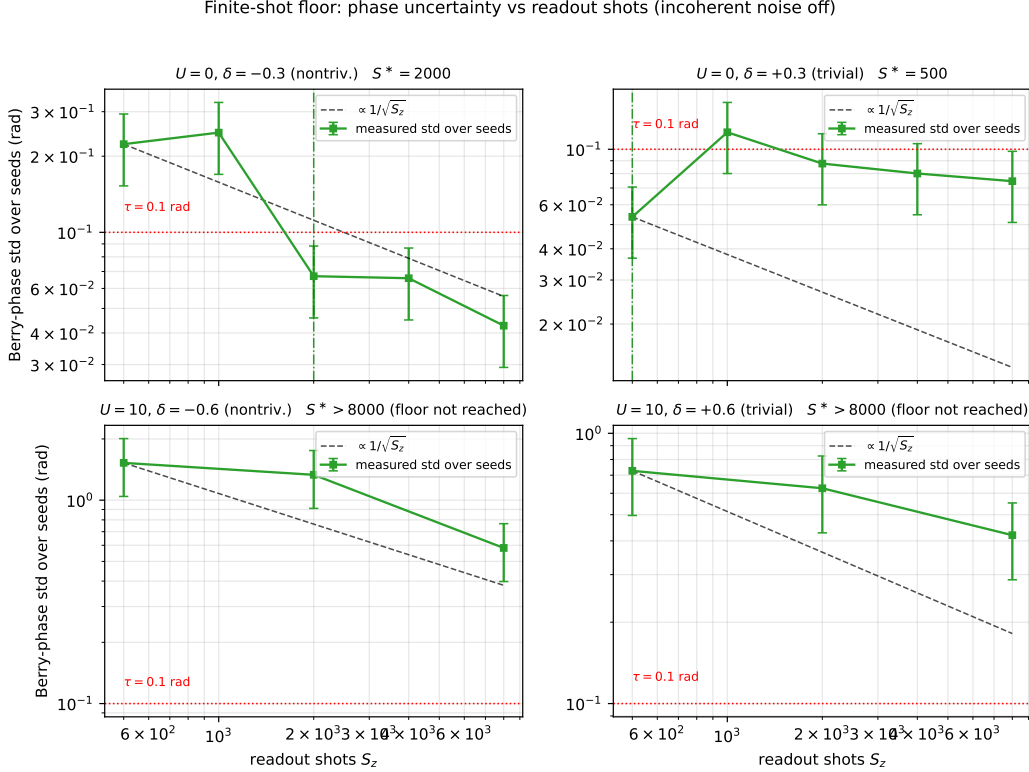


Figure 3: Finite-shot floor: seed spread (circular standard deviation over 6 seeds) of the Berry-phase estimate versus the per-leg readout shot budget S_z , with incoherent noise off (only finite-shot sampling active) and $S_{\text{ref}}=2000$ energy shots, for all four configurations. The dashed curve is a $\propto 1/\sqrt{S_z}$ reference through the lowest-shot point; the dotted line is the $\tau=0.1$ rad tolerance. The shallow $U=0$ configurations (top) cross into tolerance at $S^*=2000$ (nontrivial) and 500 (trivial); the deep $U=10$ configurations (bottom) remain above tolerance at every S_z sampled, so their floor is reported as the lower bound $S^*>8000$ — it is set not by the readout but by the fixed- S_{ref} energy-node accumulation over the much deeper circuits (see text).

For both $U=0$ configurations the seed spread falls as S_z increases and crosses the $\tau=0.1$ rad tolerance at $S^*=2000$ shots (nontrivial) and $S^*=500$ shots (trivial). Two features are visible in Fig. 3. First, the readout shot noise decreases with S_z , so the spread is largest at the low-shot end (e.g. 0.22 rad at $S_z=500$ for the nontrivial configuration). Second, the spread does not fall to zero but saturates at a floor of ~ 0.04 – 0.08 rad: the $\varphi_{G,1}$ energy leg is measured at the fixed $S_{\text{ref}}=2000$ shots per Pauli term, so its binomial error is independent of S_z and sets a constant background that dominates once the readout is well-resolved. The shots axis therefore has a clear floor S^* of only a few hundred to two thousand readout shots per leg — far below the $S_{\text{ref}}=20000$ used for the noise sweep — confirming that finite-shot readout sampling is not the limiting error source at the reference budget; the incoherent $\varphi_{G,1}$ accumulation is (Sec. 5.3). Raising S_{ref} for the energy terms, not the readout, is what would reduce the residual spread.

The deep $U=10$ circuits have no reachable readout shots floor. For the $U=10$ configurations the picture is qualitatively different: the seed spread stays far above the $\tau=0.1$ rad tolerance at every readout budget we sampled — 1.53/1.33/0.58 rad (nontrivial) and 0.73/0.63/0.42 rad (trivial) at $S_z=500/2000/8000$ — so the shots floor is not reached within the grid and we report the lower bound $S^*>8000$ for both dimerisations (Fig. 3, bottom). Decomposing the spread into its legs shows why. The readout overlap contribution $|\Delta\varphi_{\text{qc}}|$ does fall with S_z (e.g. 1.26 $\rightarrow$ 0.26 rad for the nontrivial configuration over $S_z=500 \rightarrow 8000$), but the $\varphi_{G,1}$ energy leg carries a seed spread of ~ 0.38 –

0.41 rad that is *independent of S_z* : it is the binomial error of the fixed- S_{ref} energy measurement accumulated over the far larger number of nodes and gates of the 596–916-CNOT circuits. For the deep circuits this energy-node floor alone exceeds τ , so no increase of the readout budget can bring the Berry phase within tolerance — the shots floor is governed by the energy leg, mirroring the incoherent-noise result (Sec. 5.3) that $\varphi_{G,1}$ is the bottleneck. This is the finite-shot analogue of the deep-circuit fragility seen in Fig. 1: where the shallow $U=0$ circuits need only $S^* \sim 10^{2.7} - 10^{3.3}$ readout shots, the deep $U=10$ circuits would require a prohibitively large energy-measurement budget before the readout shots even matter.

5.3 Mechanism: $\varphi_{G,1}$ dominates the degradation

The three legs of Eq. (1) respond to noise very differently, and the difference explains where the threshold sits. φ_{qc} is the argument of a single complex overlap z : device noise shrinks $|z|$ (dephasing the ancilla) but, to first order, rotates its phase only weakly, so φ_{qc} is comparatively robust. $\varphi_{G,1} = -\int \langle H \rangle dt'$ is a *dynamical-phase* leg: each node’s measured energy carries an independent stochastic error, and because the leg is a signed time integral over the loop these errors accumulate as a random walk that does *not* enjoy the forward/return time-reversal cancellation that protects the geometric legs φ_{qc} and $\varphi_{G,2}$. The two geometric legs, by contrast, co-evolve along the same variational trajectory so that their *deterministic* biases partially cancel (the paper’s Appendix-A argument); their *stochastic* fluctuations, drawn from independent measurement circuits, add in quadrature and do not cancel. We therefore expect, and confirm, that $\varphi_{G,1}$ is the dominant noise-accumulation channel and sets the threshold. The depth dependence of the threshold follows directly: the deep $U=10$ circuits carry many more noisy gates and integrate more energy nodes, so both their incoherent dephasing (visible in the faster collapse of $|z|$ and the higher $1-F$, Fig. 2) and their accumulated $\varphi_{G,1}$ error are larger at fixed s , pushing the threshold to the smallest noise we resolve.

The per-leg decomposition of the measured error bears this out quantitatively. Splitting the total folded error into the part attributable to the readout ($|\Delta\varphi_{qc}|$) and to the dynamical leg ($|\Delta\varphi_{G,1}|$), the dynamical leg dominates at every noise scale and in both regimes. For the nontrivial $U=0$ configuration $|\Delta\varphi_{G,1}|$ grows $0.037 (s=0) \rightarrow 0.13 (0.03) \rightarrow 0.37 (0.10) \rightarrow 1.03 (0.30) \rightarrow 3.00 (1.0)$ rad, while $|\Delta\varphi_{qc}|$ grows only $0.011 \rightarrow 0.040 \rightarrow 0.15 \rightarrow 0.60 \rightarrow 1.60$ rad; for the trivial configuration $|\Delta\varphi_{G,1}|$ reaches 6.67 rad at $s=1$ against 1.91 rad for $|\Delta\varphi_{qc}|$. Three consequences follow. First, the energy-measurement leg — not the Hadamard readout — is the accuracy bottleneck, so error mitigation should target the $\langle H \rangle$ estimation (term grouping, shot allocation, symmetry verification). Second, the trivial regime is the more fragile of the two *because* its dynamical-phase leg is larger ($|\varphi_{G,1}| \approx 7.3$ versus 5.4 rad at the anchor): the accumulated energy-measurement noise scales with the magnitude of the integrated energy, so the nominally “easier” (trivial, shallower) trajectory is in fact less noise-tolerant. Third, the ancilla coherence $|z|$ decays with noise ($0.236 \rightarrow 0.154$ for nontrivial $U=0$ over the swept range), reflecting ancilla dephasing, but the associated *phase* error stays secondary to the $\varphi_{G,1}$ accumulation until the highest noise levels.

5.4 The $\varphi_{G,2}$ measurement obstruction

The geometric leg $\varphi_{G,2}$ requires the derivative overlaps $w_\mu = \langle \Psi | \partial_{\theta_\mu} \Psi \rangle = \langle \Psi | D_\mu \rangle$, with $|D_\mu\rangle = -i U_n \cdots U_{\mu+1} \hat{A}_\mu |\Phi_\mu\rangle$ and $|\Phi_\mu\rangle = U_\mu \cdots U_1 |\Psi_0\rangle$. We find that w_μ is *not accessible* to the single-ancilla circuits the method provides, for a structural reason. The natural Hadamard realisation uses a controlled Pauli rotation $e^{-i\frac{\pi}{2}\hat{A}_\mu} = -i\hat{A}_\mu$ (since $\hat{A}_\mu^2 = I$); but for the even- Y Hamiltonian pool this operator is *non-unitary on the system*, so the ancilla coherence collapses onto $|\langle \Phi_\mu | \hat{A}_\mu | \Phi_\mu \rangle|$ and the test does not return w_μ . The ancilla-free alternative — measuring the local connection $g_\mu = \langle \Phi_\mu | \hat{A}_\mu | \Phi_\mu \rangle$ on the prefix state — also fails, because the tail generators U_k ($k > \mu$) do not commute with $\hat{A}_\mu$, so $\text{Im } w_\mu \neq -g_\mu$ and w_μ carries a nonzero real part through the tail.

We verified both failures numerically against the exact state-vector value `EvalCache.w`. On the $U=0$, $\delta = -0.3$ trajectory the true per-node integrand is essentially zero (e.g. $\text{Im } \sum_\mu w_\mu \dot{\theta}_\mu = 0.00000$ at the $\lambda=\pi/2$ node), whereas the ancilla-free local-connection estimator returns $+2.07$ at the same node and -2.07 later — wrong in both magnitude and sign. The single-ancilla Hadamard circuit likewise returns values unrelated to w_μ (e.g. $-0.126 - 0.301i$ versus the true -0.895 for $\mu=0$). A

faithful measurement of w_μ requires an additional ancilla register to coherently control the whole tail, which lies outside the paper’s circuit design. This is a substantive methodological finding: *within the paper’s single-ancilla architecture, the $\varphi_{G,2}$ leg cannot be measured on device*, independently of any noise model. Combined with the empirical smallness of $\varphi_{G,2}$ (Sec. 3.6), this justifies holding $\varphi_{G,2}$ at its frozen noiseless value and restricting the noisy legs to φ_{qc} and $\varphi_{G,1}$.

6 Discussion and limitations

Several scope decisions bound the interpretation of our thresholds, and we state them plainly. (i) *Frozen trajectory*. Noise acts on the measurement of the Berry-phase constituents along a noiselessly constructed variational trajectory; the McLachlan integration and ansatz growth are not re-run under noise. The threshold therefore characterises readout and integrand-measurement resilience, not the noise-driven evolution of the trajectory itself (the optional noisy-construction study). (ii) $\varphi_{G,2}$ *frozen*. As shown in Sec. 5.4, $\varphi_{G,2}$ is both negligible here and not measurable by the paper’s single-ancilla circuits; it is held at its noiseless value. The threshold is thus a whole-method number for $\varphi_{qc} + \varphi_{G,1}$, disclosed as narrower than the fully general $\varphi_{qc} + \varphi_G$. (iii) *Quadrature systematic*. The per-half trapezoid at $n=100$ nodes/half carries a small deterministic bias (measured exactly and subtracted); the reported stochastic threshold is independent of it. (iv) *Anchor deviation*. Because the short-loop tracking is not reproduced (Sec. 4), our anchors are best-effort trajectories whose noiseless Berry phases already deviate from ED; the thresholds measure degradation *relative to those anchors*, and the anchor-to-ED gap is reported separately. (v) *Computational cost*. The density-matrix simulation of the 9-qubit circuits is the bottleneck ($\sim 5\text{--}22$ s per energy node), which is why the $U=10$ deep circuits (~ 280 generators) are characterised at reduced scope.

7 Conclusion

We have characterised the hardware-noise resilience of the Mootz–Yao AVQDS Berry-phase algorithm on the SSH–Hubbard chain. The readout overlap φ_{qc} is comparatively robust; the dynamical-phase leg $\varphi_{G,1}$ is the dominant noise-accumulation channel and sets the accuracy threshold, because its measured-energy errors integrate over the loop without the time-reversal cancellation that protects the geometric legs. We have further shown that the geometric leg $\varphi_{G,2}$ is not measurable within the paper’s single-ancilla circuit design (a structural obstruction, independent of noise), and that the paper’s short-loop non-adiabatic tracking is not reproduced by a from-scratch McLachlan AVQDS, which we trace to a variational-manifold failure. These results temper the near-term promise of the shallow-circuit advantage and identify error mitigation of the energy-measurement leg as the priority for any hardware implementation. Quantitatively, the shallow $U=0$ circuits tolerate incoherent noise up to an equivalent two-qubit error of a few 10^{-5} ($s^* \approx 0.03$ nontrivial, $\lesssim 0.01$ trivial) at $S_{\text{ref}} = 20000$ shots, while the $U=10$ circuits, an order of magnitude deeper, are already past threshold at the weakest noise we resolve ($s^* < 0.01$). The finite-shot axis shows the same depth dependence: the shallow circuits need only $S^* \sim 10^{2.7}\text{--}10^{3.3}$ readout shots, whereas the deep circuits have no reachable readout shots floor ($S^* > 8000$), limited instead by the fixed-budget energy-measurement accumulation.

References

- [1] M. Mootz and Y.-X. Yao, Efficient Berry phase calculation via adaptive variational quantum computing approach, 2025, arXiv:2506.19150.
- [2] M. V. Berry, Quantal phase factors accompanying adiabatic changes, *Proceedings of the Royal Society of London A* **392**, 45–57 (1984).
- [3] J. Zak, Berry’s phase for energy bands in solids, *Physical Review Letters* **62**, 2747–2750 (1989).
- [4] R. Resta, Macroscopic polarization in crystalline dielectrics: the geometric phase approach, *Reviews of Modern Physics* **66**, 899–915 (1994).
- [5] Y.-X. Yao, N. Gomes, F. Zhang, C.-Z. Wang, K.-M. Ho, T. Iadecola, and P. P. Orth, Adaptive variational quantum dynamics simulations, *PRX Quantum* **2**, 030307 (2021).

- [6] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall, An adaptive variational algorithm for exact molecular simulations on a quantum computer, *Nature Communications* **10**, 3007 (2019).
- [7] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, Variational quantum algorithms, *Nature Reviews Physics* **3**, 625–644 (2021).
- [8] W. P. Su, J. R. Schrieffer, and A. J. Heeger, Solitons in polyacetylene, *Physical Review Letters* **42**, 1698–1701 (1979).
- [9] A. Yu. Kitaev, Fault-tolerant quantum computation by anyons, *Annals of Physics* **303**, 2–30 (2003).
- [10] G. A. L. White, K. Modi, and C. D. Hill, Filtering crosstalk from bath non-Markovianity via spacetime quantum process tomography, *PRX Quantum* **3**, 030310 (2022).